Group D: Condenser Design Algorithms

ME6220 Heat Exchanger Design, IIT Madras. For the final compilation by the Theories, Charts and Algorithms group, and for integration by the Software group.

Code, tests and full derivations: github.com/Raghav589/condenser-design-app (private; ask Group D for access).

How to read this document

- Five algorithms: condenser selection, direct-contact, plate, vertical shell-and-tube, vertical downflow air-cooled. Each lists its steps, equations, iteration, checks and how far it has been tested.
- Sources: HSB = Hewitt, Shires & Bott, Process Heat Transfer (1994). HEDH = Heat Exchanger Design Handbook. L = ME6220 lecture number.
- Every algorithm is implemented in Python as design(inputs) -> DesignResult. All numbers quoted come from running the code.
- Results are preliminary designs. Mechanical sizing is a screening calculation from HEDH Part 4 and does not establish code compliance.

Common workflow

Stage	What happens	In the code
Input	Typed inputs with unit, range and source metadata. Internal units are SI, temperatures in °C	schemas.py
Validation	Required fields, ranges and choices, then module physics checks. Bad input stops here	common/validation.py, module.validate()
Selection	HSB Fig. 17.1 chart recommends a type and a fluid allocation	modules/selection.py
Design	Module algorithm, recording each step as formula, numbers, result	module.design()
Checks	Each check is pass, fail or not assessed, tagged thermal, hydraulic, geometric or mechanical	DesignResult.check()
Report	Dashboard, Excel and PDF from the same result	reporting.py

Theory shared by the tubular algorithms

Local coefficient for vertical in-tube downflow

$\Gamma = \dot{m}(1 - x)D/4$, $Re_f = 4\Gamma/\eta_L$ [HSB Ex. 16.2, Eq. 16.19]

$\alpha^+ = (\alpha/\lambda)[\eta^2/(\rho_L(\rho_L - \rho_G)g)]^{1/3}$ [HSB Sec. 16.3.1]

$\alpha^+ = (4/3)^{1/3}\, Re_f^{-1/3}$, $Re_f < 30$; $0.756\, Re_f^{-0.22}$, 30 to 1600; $0.023\, Re_f^{0.25}\, Pr^{0.5}$, above 1600 [HEDH 2.6.2 Eq. 17; HSB Eq. 16.26]

$\tau_i = (-dp_F/dz)\, D/4$ [HSB Eq. 16.53]

$\alpha_{sh} = c_{pL}\, (\rho_L\, \tau_i)^{1/2} / T^+(\delta^+(Re_f), Pr)$ (Kosky-Staub) [HSB Eqs. 16.46-16.52]

$\alpha = (\alpha_{sh}^2 + \alpha_{grav}^2)^{1/2}$ [HSB Eq. 16.56]

Cross-check: $\alpha_{LO} = 0.021\, (\lambda_L/D)(\dot{m}D/\eta_L)^{0.8}\, Pr^{0.43}$, $\alpha = \alpha_{LO}\, [1 + x(\rho_L - \rho_G)/\rho_G]^{1/2}$ (Boyko-Kruzhilin) [HSB Eqs. 16.54-16.55]

Why Eq. 16.56: it is the method HSB gives for this geometry, and it tends to the Nusselt result at low shear and to the shear result at high shear. HSB works no example of the combination, so it is tested through limits and hand values.

Two-phase pressure drop

$(dp/dz)_F = \varphi_{LO}^2\,(dp/dz)_{LO}$, $\varphi_{LO}^2 = E + 3.24\,F\,H/(Fr^{0.045}\,We^{0.035})$ (Friedel) [HSB Eqs. 10.49-10.55]

$E = (1-x)^2 + x^2\,\rho_L\,f_{GO}/(\rho_G\,f_{LO})$, $F = x^{0.78}\,(1-x)^{0.24}$, $H = (\rho_L/\rho_G)^{0.91}\,(\eta_G/\eta_L)^{0.19}\,(1-\eta_G/\eta_L)^{0.7}$

Total = friction + acceleration + gravity, with a void fraction from Premoli [HSB Eqs. 10.26, 10.27, 10.57-10.64]

Neither book prints Friedel's downflow form, so the printed form is used, as HSB does in Ex. 16.2. HSB prints 3.24 and $(1-x)^{0.24}$, HEDH 2.3.2 prints 3.23 and $(1-x)^{0.224}$. The code uses HSB's, because its worked examples do.

Flooding, for reflux

$\dot{m}_{crit} = C^2\,[g\,D\,(\rho_L - \rho_G)]^{1/2} / (\rho_G^{-1/4} + \rho_L^{-1/4})^2$, $C = 0.75$ (HSB), $C^2 = 0.53\,(\sigma/0.072)^{0.5}\,F_2$ (HEDH) [HSB Eq. 18.34; HEDH 3.4.9C]

The design point is 75% of the lower of the two limits. They differ by 2.7 times for pentane.

1. Condenser selection and fluid allocation

Source: HSB Fig. 17.1 and Sec. 17.2; allocation rules from L23. The book describes the chart as a broad indication.

#	Question	Yes	No
1	Coolant: utility or process?	utility: 2	process: 3
2	Economic cooling water available?	3	Air-cooled
3	Plate-type exchanger feasible?	Plate	utility: 5; process: 4
4	Very clean fluid and close approach?	Plate-fin	5
5	Vapour corrosive or dirty?	6	8
6	Multicomponent with total condensation?	8	7
7	High solids, or countercurrent mandatory?	Baffle column	Spray
8	Reflux operation required?	Vertical tube-side reflux	9
9	Horizontal orientation required?	10 (horizontal)	10 (vertical)
10 V	High pressure or corrosive?	Vertical tube-side	11, then 12
12	Coolant preferred on the tube side?	Vertical shell-side	Vertical tube-side
10 H	High pressure or corrosive?	Horizontal tube-side	11
11 H	Low pressure drop required?	13	Horizontal E-shell
13	Countercurrent flow required?	Rod-baffle	14
14	Wide condensation range or venting needed?	J-shell	X-shell

Output: recommended type, the design module to run with its variant chosen, and fluid allocation advice. Plate-fin and the horizontal types are named with a warning, because no design module exists for them. Tested: each of the 13 outcomes reached by its own path.

2. Direct-contact condenser

Sources: HSB Ch. 22, HEDH 2.6.8, L23. Types: spray, jet, sheet, fan, baffle column, packed bed, bubble column, pool.

Every type: balance and outlet state

Give exactly two of: vapour flow, coolant flow, coolant outlet. The third is solved.

$Q = (M_v - M_{vv})\, h_{fg} + M_v\, c_{pv}\, (T_{v,in} - T_{sat}) + M_{vv}\, c_{pv}\, (T_{sat} - T_{vent}) + M_g\, c_{pg}\, (T_{v,in} - T_{vent})$ [HEDH Eq. 1; HSB Eq. 18.15]

$Q = M_L\, c_{pL}\, (T_{Le} - T_{Li})$, $\theta = (T_{Le} - T_{Li})/(T_{sat} - T_{Li})$ [HSB Eqs. 22.9-22.11]

$T_{mix} = (M_L\, c_{pL}\, T_{Le} + M_c\, c_{pc}\, T_{sat})/(M_L\, c_{pL} + M_c\, c_{pc})$ [First law]

Spray

#	Step	Source
1	$Ja = \rho_c\, h_{fg}/[\rho_L\, c_{pL}\, (T_{sat} - T_{Li})]$. Check Ja > 20	HEDH Eq. 13
2	$Fo = -\ln(1 - \theta^2)/\pi^2$, residence time $t = Fo\, R^2/\kappa_L$	HSB Eq. 22.9
3	Drop velocity and distance with gravity neglected: $V = (V_{in}^{-0.16} + 3.23\, \Gamma t/D^{1.84})^{-1/0.16}$, $x = 0.06\, (D^{1.84}/\Gamma)(V_{in}^{0.84} - V^{0.84})$	HSB Eqs. 22.5-22.7
4	Distance with gravity: integrate $dV/dt = g(1 - \rho_G/\rho_L) - 27\, Re^{-0.84}\, (3/4D)(\rho_G/\rho_L)\, V^2$ by RK4, doubling steps until the distance moves under 0.01 mm	HSB Eqs. 22.1, 22.4
5	$H = \max(x, x_g)$, because the drop needs the same time and travels further under gravity	This work; HSB Sec. 22.4.1.3 for the method
6	Drop growth $D'/D = (1 + \theta/Ja)^{1/3}$. Check under 3%	HSB Eq. 22.14

Design basis: one drop at the maximum size, as HEDH advises for preliminary design. Vessel cross-section needs the fraction of terminal velocity, which no book gives, so it is a design input.

Jet, sheet, fan

	θ ≥ 0.45 (small Gz)	θ < 0.45 (large Gz)
Sheet	$1 - \theta = (8/\pi^2)\, \exp(-4\pi^2/Gz)$ [HSB 22.17]	$\theta = 8/(\pi\, Gz)^{1/2}$ [HEDH 25]
Jet	$1 - \theta = 0.6915\, \exp(-23.136/Gz)$ [HEDH 34]	$\theta = 8/(\pi\, Gz)^{1/2}$ [HEDH 38]
Fan	$1 - \theta = (8/\pi^2)\, \exp(-4\pi^2/(3Gz))$ [HEDH 36]	$\theta = 8/(3\pi\, Gz)^{1/2}$ [HEDH 40]

Length: $x = D_h^2\, u/(\kappa_L\, Gz)$, $D_h = 2w$ (sheet), D (jet). The switch at 0.45 is where the asymptotes cross.

Baffle column

No stage-count procedure is published, so the count is built from sourced parts and reported as an estimate. Each fall is a sheet: $Gz = (2w)^2 u/(\kappa_L\, x_s)$, θ_s from the sheet row above. Mixed after each stage: $T_{n+1} = T_n + \theta_s\, (T_{sat} - T_n)$, marched to T_{Le}. $H = N\, x_s$.

Packed bed

$C = M_L\, c_{pL}/(M_G\, c_{pG})$, $Da = 6(1 - \varepsilon)$ for spheres, $H = (2.5\, C^{0.21}/Da)^3$, $Z = H\, D$ [HSB Eqs. 22.27, 22.33b]

$St = 0.40\, Ja\, C^{-0.21}\, H^{-0.67}$ [HSB Eq. 22.24]

With no measured U_v, HEDH's largest reported 2.79×10^6 W/m³K gives a lower bound on height. Vapour velocity is checked below about 4 m/s.

Bubble column

$Ja' = \rho_L\, c_{pL}\, (T_{sat} - T_\infty)/(\rho_G\, h_{LG})$ [HSB Eq. 22.34]

$Fo_c = 2\pi^{1/2}/(3\, K_v^{1/2}\, Ja'\, Pe^{1/2})$, $t = Fo\, D_0^2/(4\kappa_L)$, $z = v\, t$ [HSB Eq. 22.43]

Stepped collapse with v(D): HSB Eq. 22.41. Column height from HEDH Eqs. 87-88 when release frequency and spacing are given.

Pool and tail pipe

Pool: balances only, because no sizing procedure is published. Tail pipe: $H = (p_{atm} - P)/(\rho_m g)$, compared with the published 10 to 10.36 m.

Checks and testing

- $T_{sat} - T_{Le} \geq 1$ °C; drops 125-520 µm; bubble column $T_{sat} - T_{in}$ above 2 °C pure, 3.5 °C with gas (HEDH 2.6.8).
- Benchmarks: HSB Ex. 22.1-22.4. Example: duty 59.98 kW, height 2.15 m with gravity against 1.62 m without.
- Not benchmarked: jet, fan, baffle column, pool.

3. Plate condenser

Sources: HSB Sec. 8.4 and 20.2, HEDH 3.7 and 4.4.2, L13-L17. Vapour on one pass in downflow, coolant counterflow.

#	Step	Source
1	Duty $Q = \dot{m}_v h_{fg} (x_{in} - x_{out})$	L17 Step 1
2	Coolant: exactly one of outlet temperature, flow or assumed effectiveness P (0.85-0.90) sets the other two	L17 Step 1
3	P, required NTU at R = 0, LMTD	L17 Step 2; HSB 8.23, 8.41
4	Plate geometry: $D_e = 2b$ (projected) or $2b/\phi$ (developed), a, A_f, R_w	L14; HSB 8.2, 8.7, 8.8
5	For trial N (odd): channel flows, $G = \dot{m}_{ch}/(bW)$, $Re = G D_e/\mu$, $Nu = C_h Re^y Pr^n (\mu/\mu_w)^m$, $h = E_h Nu\, k/D_e$	L14, L17; HSB Ex. 8.1
6	Quality zones of equal Δx: $X^2 = (dp/dz)_L/(dp/dz)_G$; $\phi_L^2 = 1 + C/X + 1/X^2$; $\varepsilon_G = (\phi_L - 1)/\phi_L$; Akers G_{Le}; h_i; U_i; $\Delta A_i = \Delta Q/(U_i \Delta T_i F)$; $A_{req} = \Sigma \Delta A_i$	HSB Secs. 20.2.1-20.2.2
7	Iteration 1: $N_{next} = A_{req}(N)/a$, rounded up to an allowed count, from the smallest allowed N, until unchanged	L17 Steps 5-10; HSB 20.2.2
8	Pressure drops: $\Delta P_{ch} = 4f(L N_p/D_e)(G^2/2\rho)(\mu/\mu_w)^{-m}$, $\Delta P_{port} = 1.4(G_p^2/2\rho)N_p$, static, two-phase on the vapour side	L14; HSB 8.4.4, 20.2.1
9	Iteration 2, only if an allowable is exceeded: add plates and accept excess area, then step back while the limits hold	HSB Ex. 8.4
10	Checks: correlation range, area basis, pressure and temperature limits, gasket table, maldistribution indicators	See below
11	Cross-check: digitised approximate-design curves on the coolant side	HEDH 3.7.10C; L16

Built-in plate correlations

Name	Nu	f (Fanning-type)	Re	Source
HSB chevron (default)	$0.4\, Re^{0.64}\, Pr^{0.4}$	$2.78\, Re^{-0.18}$	150 to 10000	HSB Ex. 8.1, 8.2
HEDH intermating	$0.2\, Re^{0.67}\, Pr^{0.4}\, (\eta/\eta_w)^{0.1}$	$1.22\, Re^{-0.252}$	1000 to 10000	HEDH 3.7.4-3.7.5
Entered	$C_h Re^y Pr^n (\mu/\mu_w)^m$	$K_p\, Re^{-z}$	As typed	L14

Chevron angle acts through enhancement factors: 1.4 on h and 3 on f at 45°, 2 and 10 at 60°, relative to 30° (HSB Ex. 8.4).

Digitised approximate-design curves (HEDH 3.7.10 Figs 1-3)

Plate area: $a = 2.958\, V^{0.505}$ (a in m^2, V in m^3/s; read ±1.5%)

Coefficient: $\alpha = 3205\, \Delta p^{0.2986}$ (α in W/m^2K, Δp in kN/m^2; read ±1%)

Plate flow: $\log_{10} q = c + m \log_{10} \Delta p$, with (a, c, m) = (0.1, −4.1209, 0.5773), (0.25, −4.0141, 0.5843), (0.5, −3.9296, 0.5762), (1.0, −3.8883, 0.5868), (2.0, −3.8510, 0.5951)

These fits are ours, through points read from the charts. Against the readings in HEDH's Ex. 1 and 2, they differ by 0.4% to 5%.

Akers equivalent mass flux

$G_{Le} = G(1 - x) + G_e$, with $G_e = G \times (\rho_L/\rho_G)^{1/2} [(dp/dz)_G/(dp/dz)_L]^{1/2}$ evaluated at a common mass flux. HSB's wording is "local conditions", but each phase's actual flow gives absurd values and the wrong limit at x = 0. HSB's Eq. 20.7 as printed is wrong: it has x and 1 − x swapped.

Gasket limits (HEDH 4.4.2 Table 2)

Rubber, max temperature	Suitable for	Unsuitable for
SBR, 87 °C	Aqueous solutions, dilute acids	Strong acids, many organic liquids
Nitrile, 137 °C	Aqueous liquids, dilute acids, fats, mineral oils	Strong oxidizers, strong acids, polar organic solvents
Butyl, 152 °C	Aldehydes, ketones, inorganic and organic acids	Fats, oils, aromatics, concentrated mineral acids
EPDM, 152 °C	As butyl, frequently superior	Fats, oils, aromatics, concentrated mineral acids
Viton, 177 °C	Wide range of organics, fats and oils, some mineral acids	Esters, ketones
Silicone	Specific duties, low temperature	Many organic liquids

Testing

- Benchmarked: coolant side and sizing against HSB Ex. 8.1-8.4; two-phase relations against HSB Ex. 10.1, 10.3.
- Not benchmarked: the whole condensing path, because no source works a complete condensing plate example. Plate stress has no procedure in the books and stays not assessed.
- Example: Q = 225.7 kW, 15 thermal plates, U = 2901 W/m²K, LMTD = 39.9 K.

4. Vertical shell-and-tube condenser

Sources: HSB Ch. 6, 16, 18; HEDH 3.3, 3.4, 4.2, 4.3, 4.6; L9-L12, L23, L24. Variants: in-tube downflow (primary), unbaffled shell-side, reflux upflow.

#	Step	Source
1	Duty: latent + superheat + subcooling; one of $T_{c,out}$ and M_c gives the other	HSB 18.12, 18.17, 18.22
2	First guess N: 8 (fewest per pass), or A = Q/(U ΔT_m) if U is entered, or 75% of flooding for reflux	HEDH 3.3.5, 3.4.9
3	N gives D_{ctl}, D_{otl} and D_s, with L_{bb} from Fig. 14 at the bundle type; L_{bc} from the spacing ratio; L_{tb} from Fig. 12; L_{sb} = 3.1 + 0.004 D_s mm	HEDH 3.3.5 Eqs. 16-21, Figs 12-14
4	Coolant on the shell: Bell-Delaware h_c and ΔP_s; on the tubes: tube-side correlations	HEDH 3.3.6-3.3.8; L10
5	Zone 1, superheat: α_v, U_s, T_{h*}, T_{c*}, Q_{dsh}, A_{dsh}. No dry zone if $T_{h*} \geq T_{h,in}$	HSB 18.10-18.16; L24
6	Zone 3, subcooling: falling film, iterated on its length if laminar; $T_{c'}$, A_{sub}	HEDH 3.4.6G; HSB 18.20-18.23
7	Zone 2, condensing: equal Δx steps; α from Eq. 16.56 (times 1 − 0.18C with non-condensables below 5 wt%); local U; log-mean ΔT of the step; dA = dQ/(U ΔT)	HSB 16.3.3, 18.3.1, 3.17; HEDH 3.4.9B

#	Step	Source
8	$A_{req} = A_{dsh} + A_{con} + A_{sub}$; $L_{req} = A_{req}/(\pi D_o N)$	HSB 18.24
9	Condensing ΔP: friction + acceleration + gravity along the true x(z); the allowable is compared with friction, because recoveries are not credited	HSB 18.4.1, 10.27; HEDH 3.4.7
10	Compare with L, velocity limits, allowables (and 75% of flooding); update N; repeat from step 3	HEDH 3.4.9B, C
11	At the final N: cross-checks (Boyko-Kruzhilin, Chen, other suction setting, Kern), then checks	HSB 16.3.2.2, 6.3
12	Next standard shell and its tube count, reported; the design is not re-run on it	HEDH 4.2.5 Table 2a
13	Mechanical screening: shell, tube, tubesheet	HEDH 4.2.6, 4.3.6, 3.3.5
14	Vibration screening: f_n, u_{crit}, f_{vs}, f_{tb}	HEDH 4.6

The iteration on the tube count

- Why: N sets the condensing mass flux, so α, U, area and two-phase drop; through D_s it sets the shell-side velocity, h_c and drop. The area cannot be found before the count.
- First guess: low on purpose, because each update can only raise the count until the limits are met.
- Update: while a limit is exceeded and nothing has passed, N_{next} = ceil(N × worst ratio), with ratios L_{req}/L, $(\Delta P_{cond}/\text{allow})^{1/1.75}$, $\Delta P_{coolant}/\text{allow}$, u/u_{max} and $G/(0.75\ \dot{m}_{crit})$.
- Once a count has passed: bisect between the largest failed and the smallest passed count.
- Stop: they differ by one, or the update repeats.

Pass	N	D_s (m)	U (W/m²K)	A_req (m²)	L_req (m)	Worst ratio	Next N
1	8	0.142	1724	39.23	61.446	105.237	842
2	842	1.084	562	120.35	1.791	0.448	425
3	425	0.782	724	93.41	2.754	0.689	216
4	216	0.569	895	75.57	4.385	1.096	320
5	320	0.683	792	85.36	3.343	0.836	268
6	268	0.629	838	80.69	3.773	0.943	242
7	242	0.600	865	78.20	4.050	1.012	255
8	255	0.614	850	79.53	3.908	0.977	248
9	248	0.607	858	78.76	3.980	0.995	245
10	245	0.603	861	78.48	4.014	1.004	246
11	246	0.604	860	78.57	4.003	1.001	247
12	247	0.605	859	78.66	3.991	0.998	247

Bell-Delaware, coolant on the shell (HEDH 3.3.6-3.3.8)

$S_m = L_{bc}\,[L_{bb} + (D_{ctl}/L_{tp,eff})(L_{tp} - D_o)]$, $\dot{m}_s = M/S_m$, $Re = D_o\,\dot{m}_s/\mu$

$j_i = a_1\,(1.33/(L_{tp}/D_o))^{a}\,Re^{a_2}$, $a = a_3/(1 + 0.14\,Re^{a_4})$; f_i with b_1 to b_4 (Table 1)

$\alpha_i = j_i\,c_p\,\dot{m}_s\,Pr^{-2/3}\,(\mu/\mu_w)^{0.14}$, $h_c = \alpha_i\,J_c\,J_l\,J_b\,J_r$

$J_c = 0.55 + 0.72\,F_c$; $J_l = 0.44(1 - r_s) + [1 - 0.44(1 - r_s)]\,\exp(-2.2\,r_{lm})$

$J_b = \exp[-C_{bh}\,F_{sbp}\,(1 - (2\,r_{ss})^{1/3})]$; $J_r = (10/N_c)^{0.18}$, 1 above Re = 100

$\Delta P_s = \Delta P_c + \Delta P_w + \Delta P_e$, $\Delta P_c = \Delta P_{bi}\,(N_b - 1)\,R_b\,R_l$, $\Delta P_e = \Delta P_{bi}\,(1 + N_{tcw}/N_{tcc})\,R_b\,R_s$

Mechanical and vibration screening (HEDH Part 4)

$t = P\,R/(S\,E - 0.6\,P)$ [HEDH 4.3.6 Eq. 8, quoting ASME UG-27]

TEMA class R minimum shell, carbon/alloy: 330-737 mm 9.5/4.8; 762-991 11.1/6.4; 1016-2032 12.7/7.9 mm [HEDH 4.2.6]

$T = (F\,G/2)(P/S)^{1/2}$, F = 1.0 floating head, 1.25 U-tube; fixed tubesheet: rough 0.1 D_s [HEDH 4.3.6 Eq. 17; 3.3.5 Eq. 1]

$f_n = (C_n/2\pi)[E\,I/(M_e\,L^4)]^{1/2}$, $M_e = M_t + M_w + M_s$ [HEDH 4.6.2]

$u_{crit} = \beta\, f_n\, D_o\, [M_e\, \delta_o/(\rho_s\, D_o^2)]^{1/2}$ (Connors) [HEDH 4.6.4]

$f_{vs} = Sr\, u_c/D_o$, $f_{tb} = (u_c\, D_o/(p_l\, p_t))\, [3.05(1 - D_o/p_t)^2 + 0.28]$ Pass: $u_c < u_{crit}$; $f_{vs}, f_{tb} \le 0.8\, f_n$ [HEDH 4.6.4-4.6.5]

The vibration screen uses the lowest curve $C_n = 10$ and the lowest band $\beta = 2.0$ (conservative chart reads). This is a screening result, with no code compliance claimed.

Worked example: n-pentane condenser (illustrative inputs)

Quantity	Value
Vapour	5 kg/s n-pentane, 70 °C in (superheated), T_{sat} 61.85 °C, 50 °C out (subcooled), 2.27 bar
Coolant	Water 25 to 40 °C on the shell side
Tubes	25.4 × 2.11 mm, 4 m, 31.75 mm square pitch, 25% cut, spacing 0.4 D_s
Duty, coolant flow	1915.7 kW, 30.57 kg/s
Tube count, shell	N = 247, D_s = 605 mm; next standard shell 635 mm
Area	78.66 m² required, 0.0 dry, 71.76 condensing, 6.91 subcooling
Coefficients	U = 859, $h_{coolant}$ = 4754, α condensing mean = 2446 W/m²K
Pressure drop	Condensing 0.75 kPa, coolant 19.39 kPa
Cross-checks	Area: Kosky-Staub 78.7, Boyko-Kruzhilin 80.4, Chen 85.9 m²
Mechanical	Shell 1.49 mm for pressure, TEMA minimum used 9.5 mm; u_c = 1.06 m/s against u_{crit} = 6.09 m/s
Checks	35 pass, 1 fail, 10 not assessed

Testing

- Benchmarked sub-paths: Bell-Delaware against HEDH 3.3.9 (α 1249 against 1252 W/m²K; ΔP 13.79 against 13.7 kPa); Kern against HSB Ex. 6.1; shear term against HSB Ex. 16.2; mechanical thickness against HEDH 4.3.6.
- Not benchmarked: the whole path, the Eq. 16.56 combination, the zone method and the sizing loop, because no source works a complete vertical condenser example. These have hand values, limits, closure and trends.

5. Vertical downflow air-cooled condenser

Sources: HSB Ch. 9 and 19, Sec. 2.4, Sec. 16.3; HEDH 3.4.6, 3.8, 4.4.1. Approximate design with averaged tube-side conditions, per HSB Sec. 19.4.

#	Step	Source
1	$Q = M\, h_{LG}\, (x_{in} - x_{out})$	HSB Ex. 19.1
2	Air: outlet gives $M_a = Q/[c_p(T_{out} - T_{in})]$; flow gives T_{out}; velocity gives $M_a = \rho\, v_o\, S_o$	First law; HSB Ex. 9.1
3	ΔT_{LM} with F = 1, because the condensation is at nearly constant temperature	HSB Ex. 19.1; HEDH 3.8.5
4	First guess $A = Q/(U_{typ}\, \Delta T_{LM})$, $L = A/(N\, \pi\, D_r)$	HSB Ex. 19.1, Table 9.1
5	Fin geometry: A_F, A_W, A, A_T, S_{min}, S_o, σ	HSB 2.224-2.230, 2.253
6	Air side: v_{max}, Re, $Nu = 0.242\, Re^{0.658}\, (s/h)^{0.297}\, (p_1/p_2)^{-0.091}\, Pr^{1/3}$, η_f, $\alpha_{a,r} = (\eta_f\, A_F + A_W)\, \alpha_a/A_T$	HSB 2.244, 2.247, 9.3
7	Tube side: mean α_i by the chosen method, the others as cross-checks	HSB 16.54-16.56, 3.17
8	$1/U = 1/\alpha_{a,r} + R_c + (D_r/2k_t)\, \ln(D_r/D_i) + (D_r/D_i)(R_{f,i} + 1/\alpha_i)$	HSB 9.2; HEDH 3.8.5

#	Step	Source
9	Update $L_{next} = L\, A_{req}/A_T$, or the tube count; back to step 5. Stop at $\lvert\Delta L\rvert < 0.1$ mm	This work
10	Tube ΔP: friction + momentum + static gain, with x(z) from $dz \propto dx/U(x)$	HSB 10.26, 10.27, 18.4.1
11	Air $\Delta P = (K_a + n_r K_f)\, \rho\, v_{max}^2/2$; static pressure	HSB 2.221, 2.222, 2.252
12	Fan: volume, number of fans, $N = V(\Delta p_{stat} + \Delta p_{dyn})/\eta$, motor margin; or natural-draught height	HEDH 3.8.7-3.8.8
13	Noise: $L_w = c + 30 \log w + 10 \log(\Delta p_{stat}\, V/1000) - 5 \log D$	HEDH 3.8.9 Eqs. 1-3
14	Regime and applicable method at 10 stations	HEDH 3.4.6 B
15	Sonic limit at the tube inlet, when pressure is entered	HEDH 3.8.9 B
16	Checks: correlation ranges, fan diameter 1.8-4.1 m, tip speed ≤ 61 m/s, S/S_v 1.8-2.6, motor margin, noise against indicative maxima	HEDH 4.4.1, 4.8.3

Tube-side regime rule (HEDH 3.4.6 B)

$j_{g^*} = \dot{m}x/[g\, D\, \rho_G\, (\rho_L - \rho_G)]^{1/2}$: annular at or above 1.5, no-shear at or below 0.5

$F = \tau_w/[(\rho_L - \rho_G)\, g\, s]$: shear at or above 29, no-shear at or below 5

HEDH takes these limits from horizontal-tube work and states that the same criteria are assumed to apply to vertical downflow. Default method: Eq. 16.56, because it tends to the right coefficient in both limits. Boyko-Kruzhilin has no gravity term, and the check fails for it if any station is no-shear.

Testing

- Benchmarks: HSB Ex. 9.1, 19.1, 19.2, 2.11; HEDH 3.8.9 fan problem. Ex. 19.2: L = 2.619 m (book 2.62), U = 490.0 W/m²K (book 490).
- These cases are not vertical downflow: Ex. 19.2 states no orientation and uses Boyko-Kruzhilin. So the gravity term and the static head are tested by hand values only.
- HSB Ex. 19.2's air balance does not close: 5 m/s and a 40 °C outlet need 18.81 and 10.72 kg/s. The module reports both closures.
- Tube-side method spread with the balance closed: 3.10 / 2.20 / 2.01 m (Boyko-Kruzhilin / Eq. 16.56 with Chen / Eq. 16.56 with Kosky-Staub).

Scope of all five algorithms

Area	Status
Horizontal condensers, plate-fin	Named by the selection chart; no design module
Plate: several vapour passes, subcooling, desuperheating, mixtures	No procedure in HSB or HEDH
Vertical: several tube passes, cocurrent coolant, baffled shell-side condensation	No usable procedure in HSB or HEDH
Mixtures	Vertical module only, by Silver-Bell-Ghaly on an entered condensation curve
Vent sizing, fogging	Not calculated. HEDH says there are very few rules; its guidance is reported
Flanges, nozzle reinforcement, expansion joints, plate stress	Not assessed
Code compliance (ASME, TEMA, HEI, API, ISO)	Not established. Standards used only as quoted in HEDH Part 4